

Linear Algebra II

FMA_2F001_EP

Lecture Notes

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Semester: semester 3
Date: September 23, 2026

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1 Class 1: Basic Arithmetic

The arithmetic fact $1 + 1 = 2$ was shown by Russel and Whitehead. The proof operates using Peano's axioms for the natural numbers $\mathbb{N}$.

Theorem 1: Peano's Axioms for $\mathbb{N}$

- **Axiom 1:** $0 \in \mathbb{N}$.
- **Axiom 2:** There exists a successor function $S : \mathbb{N} \rightarrow \mathbb{N}$.

From these, we can define $1 = S(0)$ and $2 = S(S(0))$.

Definition 2: Addition

Addition is defined by the following rules:

1. $\forall m \in \mathbb{N} : m + 0 = m$ (Add1).
2. $\forall m, n \in \mathbb{N} : m + S(n) = S(m + n)$ (Add2).

Example 3: Computing $1 + 1$

$$1 + 1 = 1 + S(0) \stackrel{(\text{Add2})}{=} S(1 + 0) \stackrel{(\text{Add1})}{=} S(1) = 2.$$

Theorem 4: Peano's Axiom 9 (Induction)

Let P be a unary predicate. To prove $\forall m \in \mathbb{N}, P(m)$ is true, it is enough to prove:

- $P(0)$ is true.
- $\forall m : P(m) \implies P(S(m))$.

Theorem 5: Well Ordering Principle (WOP)

Every non-empty subset of $\mathbb{N}$ has a minimum.

Proposition 6: WOP implies Induction

The Well Ordering Principle can be used to prove the principle of Induction.

PROOF. Let $S = \{m \mid P(m) \text{ is not true}\}$. If $P(m)$ is not true for every m , then S is non-empty. We apply WOP to S , meaning there is a minimum element $m_0 = \min S$. But $m_0 \neq 0$, so $m_0 = S(m)$ for some m . This creates a contradiction. $\square$

1.1 Relations on a Set

Definition 7: Relations and Equivalence

Let S be a set with elements $x, y \in S$. A relation on S is defined as a subset of $S \times S$. An equivalence relation must satisfy all three of the following properties:

- **Reflexivity:** $\forall x \in S, x \sim x$.
- **Symmetry:** $\forall x, y \in S, x \sim y \iff y \sim x$.
- **Transitivity:** $\forall x, y, z \in S, x \sim y \text{ and } y \sim z \implies x \sim z$.

Example 8: Equivalence Relations

“Same Parity” and “Same remainder mod n ” are equivalence relations.

Remark 9: Partitions

An equivalence relation is the same as a partition of S .

Definition 10: Modular Classes

For every $n \in \mathbb{N}$, $a \equiv b \pmod{n}$ defines an equivalence relation. The set of equivalence classes for this relation is denoted $\mathbb{Z}/n\mathbb{Z}$.

1.2 Modular Arithmetic and Divisibility

Theorem 11: Chinese Remainder Theorem (Sunzi)

If m and n are coprime, there is a bijection between $\mathbb{Z}/mn\mathbb{Z}$ and $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

Example 12: Applying the Chinese Remainder Theorem

For $n = 2$ and $m = 3$, we can map between $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. If we want to find all $m \in \mathbb{Z}$ such that $m \equiv 2 \pmod{5}$ and $m \equiv 3 \pmod{7}$, the solution is the set $\{14 + 35k \mid k \in \mathbb{Z}\}$.

Theorem 13: Euclidean Division

$\forall a, \forall b \neq 0, \exists (q, r)$ such that $a = bq + r$ and $0 \leq r < |b|$.

Theorem 14: Bézout's Theorem

$\exists (u, v) \in \mathbb{Z}^2$ such that $au + bv = \gcd(a, b)$.

Example 15: Euclid's Algorithm

Applying the algorithm to find coefficients for 5 and 7:

- $7 = 1 \times 5 + 2.$
- $5 = 2 \times 2 + \gcd(5, 7).$
- $1 = 5 - 2 \times 2 = 5 - 2 \times (7 - 5) = 3 \times 5 - 2 \times 7.$

We have found that $3 \times 5 \rightarrow (0, 1)$ and $-2 \times 7 \rightarrow (1, 0)$. If we want to find m such that $m \equiv 2 \pmod{5}$ and $m \equiv 3 \pmod{7}$, we compute $(3 \times 5) \times 3 + (-2 \times 7) \times 2$.

1.3 Prime Numbers

Definition 16: Prime Number

A prime number is a number $p \in \mathbb{N}$ such that $p \geq 2$ and p has only divisors 1 and p .

Theorem 17: Infinitude of Primes

There are infinitely many primes.

Theorem 18: Tao's Theorem

At least one of the following sets of primes (p, q) is infinite:

- **Twin primes:** $|p - q| = 2.$
- **Cousin primes:** $|p - q| = 4.$
- **Sexy primes:** $|p - q| = 6.$

Remark 19: Goldbach Conjecture

The Goldbach conjecture posits that every even number is equal to $p + q$ with p and q primes.

2 Lecture 2: Groups

Definition 1: Group

A group is a set G equipped with a binary operation (or group law) $*$, denoted $(G, *)$, that satisfies the following properties:

1. **Associativity:** $\forall a, b, c \in G, (a * b) * c = a * (b * c).$
2. **Neutral Element:** $\exists e \in G$ such that $\forall a \in G, a * e = e * a = a.$
3. **Inverse Element:** $\forall a \in G, \exists b \in G$ such that $a * b = b * a = e.$

Optional: If a group also satisfies commutativity (i.e., $\forall a, b \in G, a * b = b * a$), it is called an **Abelian group**.

Example 2: Groups in Modular Arithmetic

- The set $\mathbb{Z}/n\mathbb{Z}$ with addition is a group. The neutral element is $[0]_n$ and the inverse of $[k]_n$ is $[-k]_n$.
- If p is prime, the non-zero elements modulo p under multiplication form a group. To find the inverse of k modulo p , one can use Bézout's identity on k and p . Because p is prime, there exist integers u and v such that $uk + vp = 1$. Modulo p , this simplifies to $uk \equiv 1 \pmod{p}$, which implies that u is the multiplicative inverse k^{-1} .

Definition 3: Powers and Cyclic Groups

For an element $g \in G$, its powers are defined as follows:

- For $n > 0$, $g^n = g * g * \cdots * g$ (n times).
- For $n = 0$, $g^0 = e$.
- For $n < 0$, $g^n = g^{-1} * g^{-1} * \cdots * g^{-1}$ ($|n|$ times).

A finite group G is called **cyclic** if it is generated by a single element g , meaning $G = \langle g \rangle = \{g^m \mid m \in \mathbb{Z}\}$.

Example 4: Common Groups

- **Dihedral groups** D_k : the group of symmetries of a regular k -gon.
- **Symmetric groups** Σ_n (or S_n): the group of permutations of n elements.
- **Alternating group** A_n : the subgroup of Σ_n containing all even permutations, i.e., $A_n = \{\sigma \in \Sigma_n \mid \varepsilon(\sigma) = 1\}$.
- The group of symmetries of polyhedra, such as the Rhombic Triacontahedron.

Definition 5: Subgroups and Morphisms

- A **subgroup** is a subset of G that forms a group under the same group law.
- A **morphism** (or homomorphism) is a map φ between groups that preserves the group structure, e.g., $\varphi(a * b) = \varphi(a) * \varphi(b)$.
- An **isomorphism** is a bijective group morphism.

Definition 6: Order of an Element

The order of an element $g \in G$ is the smallest positive integer m such that $g^m = e$. It is denoted $\omega(g) = \min\{m \geq 1 \mid g^m = e\}$.

Theorem 7: Weak Lagrange Theorem

Let G be a finite Abelian group, and let $g \in G$. Then the order of g divides the size of the group, i.e., $\omega(g) \mid \#G$.

PROOF. Let $\pi = \prod_{h \in G} h$ be the product of all elements in G . Consider the map $\varphi_g : G \rightarrow G$ defined by $h \mapsto g * h$. This map is a bijection, as it has an inverse $\varphi_{g^{-1}}$. Because G is Abelian, we can rearrange the product:

$$\pi = \prod_{h \in G} h = \prod_{h \in G} \varphi_g(h) = \prod_{h \in G} (g * h)$$

Factoring out g for each of the $\#G$ elements, we get $\pi = g^{\#G} \cdot \prod_{h \in G} h = g^{\#G} \cdot \pi$. By canceling π , we find $g^{\#G} = e$.

We can write $\#G = q \cdot \omega(g) + r$ using Euclidean division, with $0 \leq r < \omega(g)$. Substituting this into our equation gives:

$$e = g^{\#G} = g^{q\omega(g)+r} = (g^{\omega(g)})^q \cdot g^r = e^q \cdot g^r = g^r$$

Since $g^r = e$ and $0 \leq r < \omega(g)$, the minimality of the order implies $r = 0$. Therefore, $\omega(g)$ divides $\#G$. $\square$

Theorem 8: Strong Lagrange Theorem

If G is a finite group and $H \subset G$ is a subgroup, then $\#H$ divides $\#G$.

PROOF. We say two elements $x, y \in G$ are equivalent modulo H ($x \sim y$) if $y^{-1}x \in H$. This condition is equivalent to $x \in yH$, where yH represents a left coset. The left cosets form a partition of the group G , so $G = \bigsqcup_{y_i \in I} y_i H$ for some index set I of representatives.

Every coset $y_i H$ has exactly the same size, which is equal to $\#H$. Thus, we can conclude that the total number of elements in G is given by $\#G = \#H \times \#I$. This immediately shows that $\#H \mid \#G$. $\square$

Theorem 9: Cauchy's Theorem

Let G be a finite group and let p be a prime number such that $p \mid \#G$. Then there exists an element $g \in G$ such that $\omega(g) = p$.

PROOF. Consider the set $S_p = \{(g_1, g_2, \dots, g_p) \mid g_i \in G, g_1 g_2 \dots g_p = e\}$. We define an equivalence relation on S_p using cyclic shifts: two tuples are equivalent if one is a cyclic permutation of the other (i.e., shifting the sequence by some offset k). Notice that the size of S_p is $(\#G)^{p-1}$, because the first $p-1$ elements can be chosen freely, and the final element g_p is uniquely determined as $(g_1 \dots g_{p-1})^{-1}$.

Under this equivalence relation, every equivalence class contains either 1 or p elements. The tuple $(e, \dots, e)$ forms an equivalence class of size 1. We can partition S_p into these classes, yielding the equation $\#S_p = N_1 + pN_p$, where N_1 is the number of classes of size 1, and N_p is the number of classes of size p .

Since $p \mid \#G$, we know $p \mid (\#G)^{p-1}$, so $p \mid \#S_p$. Thus, $p \mid (N_1 + pN_p)$, which implies $p \mid N_1$. We already know $N_1 \geq 1$ because of the class containing $(e, \dots, e)$. Because $p \mid N_1$, we must have $N_1 \geq p \geq 2$.

This means there is at least one other class of size 1, which corresponds to a tuple $(g, g, \dots, g) \in S_p$ for some $g \neq e$. Since this tuple is in S_p , it must satisfy $g^p = e$. Because p is prime and $g \neq e$, the order of g is exactly p , so $\omega(g) = p$. $\square$